

KijaniKiosk Project Scope

KijaniKiosk capstone - project scope

Problem statement

KijaniKiosk processes payments in Kubernetes but had no automated path from a completed payment to downstream receipt processing and analytics. Receipts were not flowing into an observable, event-driven pipeline, so operations could not verify payment volume or failures without manual log inspection.

Track

Track B - Serverless-first, with Kubernetes integration as the seam between the payments service and the serverless receipt chain.

In scope (five items)

1. Four-function serverless receipt chain (kk-processor, kk-notifier, kk-analytics, plus K8s kk-payments as the event producer).
2. kk-payments on Kubernetes writing receipt JSON to environment-specific S3 buckets.
3. Jenkins pipeline with staging validation, smoke tests, and a production approval gate that records a reason.
4. Repository documentation and architecture diagram so a new engineer can run offline validation in one day.
5. AI governance log and peer-review feedback with at least one documented improvement.

Out of scope

1. Multi-region AWS failover - adds cost and complexity beyond the course lab; single us-east-1 region is sufficient for the demo.
2. Managed PostgreSQL in cloud - Week 9 used in-cluster Postgres references only; provisioning RDS is deferred to a production hardening phase.

Success criteria (measurable)

Criterion How to demonstrate

npm test and ./scripts/validate-all.sh pass Terminal output in demo or CI

serverless package succeeds for staging and production Jenkins build log or local run

Production approval gate requires APPROVAL_REASON Jenkins screenshot with input field

POST /payments writes to correct S3 bucket per overlay kubectl describe configmap + S3 object key

Serverless chain produces analytics log line CloudWatch log after receipt upload (when AWS active)

Architecture

See architecture.md and docs/architecture.png (export from Mermaid).